

Semester Group Assignment Rubric

Purpose

This rubric evaluates the semester group assignment described in:

- `big_data_course_content/semester_group_assignment_brief.md`

The assignment is a cumulative technical project. Teams are expected to build a coherent data product from ingestion to representation, clustering, recommendation or ranking, graph analysis, reproducible delivery, and final defense.

The grading priority is:

1. ability to defend decisions under questioning
2. technical correctness
3. reproducibility
4. interpretation and discussion
5. depth of analysis
6. coherence across the full project

Each assessed deliverable is graded on a **20-point** scale. If the instructor combines these deliverables into one final project grade, the final defense and question-response component should receive the heaviest external weight because the project must be defended as an integrated technical system, not only submitted as files.

Although grades may be assigned individually, the project must be defended by the group during the scheduled presentation. The group is responsible for defending the submitted work, the presentation, and the technical decisions behind the project. If the group does not defend the work in the presentation, the deliverable receives no grade.

Scoring Model

Each deliverable below has a maximum score of **20 points**. The point values inside each section sum to 20.

Assessed deliverable	Maximum score
Week 3: Dataset Charter and Processed Dataset V1	20
Week 5: Representation and Dimensionality Report	20
Week 7: Clustering and Validation Report	20
Week 10: Recommendation, Ranking, or Predictive Decision Engine	20
Week 12: Graph Analytics and Centrality Report	20
Week 14: Final Integrated Delivery	20
Final Technical Defense and Question Response	20

Minimum completeness rule:

- A project missing a real processed dataset, a rigorous feature representation, a clustering experiment, a recommendation/ranking experiment, a graph analysis, or a reproducible build path may be capped at a passing-minimum or marked incomplete, even if other sections are polished.

Performance Levels

Use these levels consistently across all components.

Level	Meaning
Excellent	Technically correct, reproducible, well justified, deeply interpreted, and clearly connected to the product question.
Good	Mostly correct and reproducible, with reasonable interpretation and only minor gaps.
Satisfactory	Basic deliverable is present, but justification, rigor, reproducibility, or interpretation is shallow.
Weak	Deliverable is incomplete, hard to reproduce, weakly connected to the project, or mostly descriptive.
Missing	Required artifact or explanation is absent.

Level	Meaning
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Defense Question Emphasis

Across all deliverables, defense questions are graded primarily on the quality of the team's reasoning. The instructor evaluates why decisions were made more heavily than simply what was done or how it was implemented. Strong answers connect evidence to tradeoffs, alternatives considered, assumptions, limitations, and the product question.

Teams are expected to take a critical view of their own work. They should identify what worked, what did not work, what could have been better, what should not be overclaimed, and what they would change with more data, more time, or a different technical constraint. This critical self-assessment is part of the defense-question score for every deliverable.

Week 3 Rubric: Dataset Charter and Processed Dataset V1

Total: 20 points

Objective:

- prove that the team has a valid domain, a valid data source, and a reproducible first dataset build

Required Deliverables and Rubric

Deliverable	Points	Excellent evidence
Project proposal	1.25	Domain, problem statement, expected product question, and course fit are specific and technically feasible.
Source inventory	1.25	Source URLs, licenses/access conditions, raw formats, estimated size, and provenance are complete and auditable.
Schema draft	1.25	Entity tables, keys, joins, grain, and relationship assumptions are clear.
Processed dataset V1	1.5	At least one cleaned table is saved to disk and can be regenerated.
Data dictionary draft	1.0	Columns, types, meanings, units, and known quality issues are documented.
Scale analysis	1.0	Rows, columns, missingness, sparsity, memory estimate, and subset justification are reported.
Ethics and Access Note	1.25	Data permission, personal-data risks, and mitigation steps are explicit.
Reproducible ingestion command	1.5	The ingestion path runs from a documented command without hidden notebook state.
Defense questions	10.0	The team can orally defend source legitimacy, schema design, data quality, scale, ethics, and how the dataset supports later course layers.

Defense Questions

Defense answers are evaluated more on why the team made each decision than on merely describing what was done or how it was implemented. Strong answers include critical self-assessment: limitations, alternatives, weak assumptions, what could have been better, and what should not be overclaimed.

- What is the exact product question, and why does this dataset support it?
- What is the grain of each table, and what are the key joins?
- Which source is highest risk, legally or technically, and how did you reduce that risk?
- What evidence shows that the processed dataset was generated reproducibly?
- What later project layers are feasible from this dataset: representation, clustering, recommendation/ranking, and graph analysis?

Interpretation and Discussion Expectations

Strong Week 3 submissions explain why the dataset can support the second half of the course. They do not only say "we have data." They explain what can become an interaction layer, a graph layer, and a recommendation or ranking task.

Common Deductions

- vague product question
- missing source permissions
- no clear entity grain
- processed data created manually without a reproducible command
- data dictionary that only repeats column names
- ethics note that says "no issue" without analysis

Week 5 Rubric: Representation and Dimensionality Report

Total: 20 points

Objective:

- move from raw tables to meaningful representations

Required Deliverables and Rubric

Subject	Excellent evidence	Max points
Feature matrix or matrices	Numeric, text, categorical, temporal, or graph-derived features are built with clear preprocessing choices.	2
Dimensionality-reduction report	PCA and/or SVD are correctly applied; optional t-SNE is clearly labeled as visualization, not compression.	1.5
Comparison table	Explained variance, retained energy, reconstruction error, or equivalent metrics are reported and interpreted correctly.	1.5
Visualization set	At least two meaningful plots support the analysis and are not merely decorative.	1
Technical interpretation	The team explains dimensionality, redundancy, feature quality, representation limits, and what each method preserves.	3
Reproducible feature-building pipeline	Feature generation can be rerun from scripts or documented commands.	1
Defense questions	The team can defend preprocessing choices, representation quality, dimensionality metrics, visualization limits, and why the representation supports the product question. Focus is more on the why rather than the what.	10

Defense Questions

Defense answers are evaluated more on why the team made each decision than on merely describing what was done or how it was implemented. Strong answers include critical self-assessment: limitations, alternatives, weak assumptions, what could have been better, and what should not be overclaimed.

- Why did you choose this representation instead of a simpler one?
- What preprocessing choice most changed the representation?
- What does PCA, SVD, or the chosen dimensionality method optimize?
- What should not be concluded from your visualizations?
- What evidence supports your interpretation beyond the plot?

Interpretation and Discussion Expectations

This milestone must answer:

- Which representation is most appropriate for the product question?
- What preprocessing choices most affected the representation?
- What dimensionality-reduction method was used, and what objective does it optimize?
- What evidence supports the interpretation beyond a visual plot?
- What are the limitations of the chosen representation?

Deep analysis should distinguish:

- PCA/SVD as compression or reconstruction-oriented methods

- text vectors as sparse lexical representations
- learned embeddings as objective-dependent representations
- t-SNE as exploratory visualization, not a production feature space

Common Deductions

- using t-SNE as proof of clusters
- reporting plots without explaining what structure is preserved
- no reconstruction or retained-energy discussion
- feature leakage or unexplained preprocessing
- no comparison against a simple baseline representation

Week 7 Rubric: Clustering and Validation Report

Total: 20 points

Objective:

- segment the domain and validate whether the segmentation is meaningful

Required Deliverables and Rubric

Deliverable	Points	Excellent evidence
K-means experiment	1.5	K values are explored, initialization is controlled, and assumptions are discussed.
DBSCAN or justified density method	1.5	Parameter choices are explored and linked to density assumptions.
Validation table	2.0	Silhouette, inertia or density-related metrics, and interpretation limits are reported clearly.
Cluster-profile analysis	2.0	Clusters are characterized with domain features, examples, and meaningful differences.
Failure analysis	1.5	The team explains what did not cluster well and why.
Parameter sweeps	1.5	Results are not based on one hand-picked run; sensitivity is shown.
Defense questions	10.0	The team can defend representation choice, clustering assumptions, parameter sensitivity, validation limits, cluster meaning, and failure cases.

Defense Questions

Defense answers are evaluated more on why the team made each decision than on merely describing what was done or how it was implemented. Strong answers include critical self-assessment: limitations, alternatives, weak assumptions, what could have been better, and what should not be overclaimed.

- Why did you cluster this representation?
- What assumptions does K-means make, and where do they fail here?
- How did DBSCAN or the density method change the interpretation?
- Which cluster is most defensible, and which is weakest?
- What validation evidence supports the cluster profiles?

Interpretation and Discussion Expectations

Strong submissions do not ask only "which metric is highest?" They discuss whether the clusters make sense for the domain and product question.

They should explain:

- which representation was clustered
- why that representation was chosen
- whether clusters are stable under parameter changes
- what each cluster means operationally
- what claims the validation metrics do and do not support

Common Deductions

- one clustering run with no sweep
- cluster labels reported with no profile
- using t-SNE islands as cluster validation
- no discussion of noise points or outliers
- no connection between clusters and the final product goal

Week 10 Rubric: Recommendation, Ranking, or Predictive Decision Engine

Total: 20 points

Objective:

- connect the technical work to a decision or ranking task

Required Deliverables and Rubric

Deliverable	Points	Excellent evidence
Baseline system	1.5	A simple baseline is implemented and justified, such as popularity-based or content-based ranking.
Stronger system	2.0	Collaborative filtering, matrix factorization, hybrid ranking, predictive ranking, or another advanced model is correctly implemented.
Offline evaluation report	2.5	Metrics, candidate-pool definition, train/test logic, and baseline comparison are explicit.
Error analysis	1.5	Strong cases and failure cases are examined with examples and domain interpretation.
Task framing explanation	1.0	The team clearly states whether the system is recommendation, ranking, prediction, or segmentation feeding ranking.
Data alignment documentation	1.5	Interaction, catalog, feature, and model inputs are aligned without hidden assumptions.
Defense questions	10.0	The team can defend the task framing, baseline, stronger model, candidate pool, metrics, error cases, and data alignment.

Defense Questions

Defense answers are evaluated more on why the team made each decision than on merely describing what was done or how it was implemented. Strong answers include critical self-assessment: limitations, alternatives, weak assumptions, what could have been better, and what should not be overclaimed.

- What is the exact candidate pool?
- Why is your baseline fair?
- What does your stronger system improve, and where does it fail?
- What metric did you optimize, and why is it appropriate for the product question?
- How did you prevent leakage across users, items, time, or interactions?

Interpretation and Discussion Expectations

The report must answer:

- What is the decision being supported?
- What is the unit being ranked or recommended?
- What is the candidate pool?
- What does a correct recommendation mean in this domain?
- Which baseline must the advanced method beat?
- Where does the model fail and why?

Depth is shown through comparison, not model complexity alone. A complex model that is not better than a simple baseline must be discussed honestly.

Common Deductions

- no baseline
 - metrics reported without candidate-pool definition
 - train/test leakage
 - no examples of good and bad recommendations
 - claiming a hybrid model without explaining data alignment
 - optimizing a metric that does not match the product question
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Week 12 Rubric: Graph Analytics and Centrality Report

Total: 20 points

Objective:

- formalize the graph induced by the domain and use it for structural analysis

Required Deliverables and Rubric

Deliverable	Points	Excellent evidence
Graph definition	2.0	Nodes, edges, weights, directionality, and graph grain are precise and justified.
Graph-construction script	1.5	The graph can be regenerated from processed data.
Graph report	2.0	Connected components, degree or weighted degree, centrality, PageRank, or related measures are computed correctly.
Comparison section	1.5	Graph ranking is compared against popularity, model-based ranking, or another baseline.
Interpretation note	2.0	The team explains what graph structure means in the domain and what it does not mean.
Validity checks	1.0	The team checks edge sparsity, isolated nodes, component structure, and sensitivity to graph-definition choices.
Defense questions	10.0	The team can defend node and edge definitions, weights, directionality, centrality meaning, comparison baselines, and graph limitations.

Defense Questions

Defense answers are evaluated more on why the team made each decision than on merely describing what was done or how it was implemented. Strong answers include critical self-assessment: limitations, alternatives, weak assumptions, what could have been better, and what should not be overclaimed.

- What does a node represent, and what does an edge represent?
- Why is the graph weighted or unweighted?
- Why is the graph directed or undirected?
- What does centrality or PageRank mean in this domain?
- How does graph ranking differ from popularity or model-based ranking?

Interpretation and Discussion Expectations

Strong graph analysis explains why this graph is the right representation. It should answer:

- What does an edge mean?
- Why is the edge weighted or unweighted?
- Why is the graph directed or undirected?
- What does centrality mean in this domain?
- How does graph ranking differ from popularity or model ranking?

Common Deductions

- graph built only for decoration

- unclear node or edge meaning
- no justification of edge threshold
- centrality reported without interpretation
- no comparison to a non-graph ranking
- no discussion of disconnected components

Week 14 Rubric: Final Integrated Delivery

Total: 20 points

Objective:

- deliver the complete system and prepare it for defense

Required Deliverables and Rubric

Deliverable	Points	Excellent evidence
Final technical report	1.5	Report follows the required structure and integrates all project layers coherently.
Reproducible repository	1.5	Repo structure is clean; raw, processed, code, reports, and artifacts are separated.
Runbook	1.5	A reader can rebuild outputs from documented steps and commands.
Final presentation	1.0	Slides communicate the system, evidence, and decisions clearly without hiding technical details.
Final demo artifact	1.5	Notebook, CLI, dashboard, or API works and shows a meaningful end-to-end use case.
Monitoring or operationalization plan	1.0	Plan discusses serving assumptions, model/data drift, retraining, logging, and failure modes.
Limitations and future work	1.0	Limitations are honest, specific, and tied to evidence from the project.
Final processed artifacts and outputs	1.0	Final data, models, reports, figures, and outputs are saved and easy to inspect.
Defense questions	10.0	The team can defend how the final report, repo, runbook, demo, limitations, and monitoring plan form one coherent system.

Defense Questions

Defense answers are evaluated more on why the team made each decision than on merely describing what was done or how it was implemented. Strong answers include critical self-assessment: limitations, alternatives, weak assumptions, what could have been better, and what should not be overclaimed.

- How do we rerun the full project from raw or staged data to final outputs?
- What is the strongest result, and what evidence supports it?
- What is the weakest part of the system, and what would you improve first?
- What would you monitor if the system were run repeatedly?
- How do the data, representation, model, graph, and demo fit together?

Final Report Alignment

The final report should include:

1. problem statement
2. domain context
3. dataset sources and access conditions
4. schema and data dictionary
5. preprocessing and feature engineering
6. dimensionality and representation analysis
7. clustering analysis
8. recommendation or ranking system
9. graph analytics
10. evaluation protocol

11. pipeline and reproducibility
12. ethics and limitations
13. final conclusions

Interpretation and Discussion Expectations

The final delivery must synthesize the semester. It should not read like separate weekly homework stapled together.

Strong submissions explain:

- why the data supports the product question
- how the representation choices affect downstream results
- what clustering revealed and what it failed to reveal
- why the recommendation or ranking system is credible
- what the graph analysis adds beyond the model
- how reproducibility was verified
- what limitations remain

Common Deductions

- final report omits one major project layer
- demo does not run
- runbook depends on undocumented local paths
- final presentation overclaims results
- monitoring plan is generic and not tied to the system
- future work is vague

Final Technical Defense and Question Response Rubric

Total: 20 points

This is the heaviest component when the instructor combines deliverables into the final project grade. Within this 20-point section, live question response is the most important evidence that the team actually understands and can defend the technical system they built.

The team must be ready for live questions about data, modeling, evaluation, graph construction, reproducibility, ethics, limitations, and product coherence.

Defense Components

Component	Points	Excellent evidence
System coherence defense	2.0	The team clearly explains how objective, data, schema, features, models, graph, evaluation, and demo form one coherent system.
Data and pipeline questions	1.25	The team can explain ingestion, cleaning, joins, data dictionary, missingness, scale, ethics, and reproducibility without handwaving.
Representation and interpretation questions	1.25	The team can defend feature choices, dimensionality methods, embeddings, visualizations, interpretation limits, and representation tradeoffs.
Modeling and evaluation questions	1.25	The team can justify baselines, advanced models, metrics, candidate pools, train/test design, and failure cases.
Graph analytics questions	1.0	The team can defend node/edge definitions, weights, directionality, centrality/PageRank interpretation, and graph ranking comparisons.
Deep analysis and discussion under follow-up questions	2.25	The team gives nuanced answers, recognizes tradeoffs, connects evidence across milestones, and avoids superficial claims.
Limitations, ethics, and operational-risk questions	1.0	The team gives honest, specific limitations; explains ethics/access constraints; and discusses monitoring or operational risks.
Individual question-response quality	10.0	Multiple team members answer clearly, technically, and directly; answers are grounded in artifacts and evidence. This is the highest-weighted line item in the defense.

Defense Questions

Defense answers are evaluated more on why the team made each decision than on merely describing what was done or how it was implemented. Strong answers include critical self-assessment: limitations, alternatives, weak assumptions, what could have been better, and what should not be overclaimed.

The defense may include questions such as:

Project Coherence

- What is the exact product question?
- What decision does your system support?
- Why does your dataset support that question?
- Which result is most important, and why?
- Which result should not be overinterpreted?

Data and Reproducibility

- What is the grain of your main entity table?
- What are the primary keys and join keys?
- What were the highest-risk data quality issues?
- How do we rerun your pipeline from raw data to final artifacts?
- Which output would change if the raw data were refreshed?

Representation and Interpretation

- Why did you choose this feature representation?
- What preprocessing choices most affected your results?
- What did PCA, SVD, embeddings, or visualizations reveal?
- What did they not reveal?
- What would be invalid to conclude from your plots?

Clustering

- Why did you cluster this representation?
- How did K-means and DBSCAN differ in your project?
- Which parameters were most sensitive?
- What evidence supports your cluster interpretation?
- Which clusters are weak or unstable?

Recommendation, Ranking, or Prediction

- What is your baseline?
- What is your stronger model?
- What is the candidate pool?
- What metric did you optimize, and why?
- Where does the model fail?
- What would happen if the data distribution changed?

Graph Analytics

- What are nodes and edges?
- What does edge weight mean?
- Why is the graph directed or undirected?
- What does centrality mean in your domain?
- How does graph ranking differ from popularity or model ranking?

Ethics and Operationalization

- Why are you allowed to use this data?
- What personal-data risks exist?
- How did you reduce those risks?

- What should be monitored if this system runs repeatedly?
- What is the biggest operational failure mode?

Defense Performance Levels

Level	Description
Excellent	Answers are precise, evidence-based, technically grounded, and integrated across the project. The team acknowledges uncertainty without collapsing the argument.
Good	Answers are mostly correct and connected to artifacts, with some gaps in nuance or cross-layer integration.
Satisfactory	Answers show basic familiarity but rely on generic explanations or shallow interpretation.
Weak	Answers are vague, inconsistent, or disconnected from the submitted artifacts.
Missing	The team cannot answer the question or no relevant artifact exists.

Defense Standards

Strong defense answers should:

- refer to specific files, tables, figures, scripts, metrics, or artifacts
- explain why a decision was made, not only what was done
- compare against at least one alternative
- identify limitations honestly
- separate evidence from speculation
- avoid claiming that a visualization alone proves a conclusion
- show that each team member understands more than their narrow task

Weak defense answers usually:

- describe tools without explaining objectives
- say "the model performed well" without a baseline
- cite plots without interpretation
- ignore data leakage or candidate-pool design
- treat graph centrality as automatically meaningful
- cannot reproduce commands
- cannot explain ethical access or personal-data risk

Cross-Cutting Rubric: Interpretation, Discussion, and Depth

These expectations apply to every milestone and final deliverable.

Interpretation Quality

Excellent interpretation:

- states what was learned
- states what was not learned
- explains why the evidence supports the claim
- connects the result to the product question
- identifies confounders, artifacts, and alternative explanations
- avoids overclaiming

Weak interpretation:

- describes a plot without explaining it
- reports a metric without context
- says a result is "good" without a baseline
- treats technical output as self-explanatory
- ignores uncertainty or limitations

Discussion Quality

Excellent discussion:

- compares alternatives
- explains tradeoffs
- links choices across pipeline stages
- uses examples from the data
- acknowledges failure cases
- proposes realistic next steps

Weak discussion:

- summarizes steps chronologically without analysis
- repeats code outputs
- avoids explaining bad results
- gives generic future work

Depth of Analysis

Excellent depth:

- includes parameter sweeps, ablations, baselines, or sensitivity checks
- shows failure analysis
- connects data quality to model behavior
- examines both aggregate metrics and concrete examples
- explains why results matter for the domain

Weak depth:

- uses a single default model run
 - skips baselines
 - has no error analysis
 - has no examples
 - treats outputs as final because the code ran
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Suggested Grading Workflow for Instructor

1. Check minimum completeness first.
 2. Review milestone artifacts against the required deliverables.
 3. Verify that the repository can be run from documented commands.
 4. Read interpretation and limitation sections before assigning high marks.
 5. During final defense, ask at least one question per project layer:
 - data
 - representation
 - clustering
 - recommendation/ranking
 - graph
 - reproducibility
 - ethics
 6. Use the final defense score to reward genuine understanding and penalize shallow ownership.
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Student-Facing Summary

The strongest projects will not simply show many models. They will show a coherent system.

To score highly, your team must demonstrate:

- a valid and ethical dataset
- a reproducible data pipeline

- meaningful feature representations
- rigorous clustering and validation
- a recommendation, ranking, or predictive decision engine with baselines
- a justified graph analysis
- clear interpretation and discussion
- honest limitations
- a working final demo
- a strong oral defense under technical questions

Each deliverable is graded out of 20. When the instructor combines deliverables into the final project grade, the final defense should carry the largest weight because it tests whether the team can explain and defend the project as engineers and analysts, not only submit artifacts.